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Benchmarking optimization methods for materials research: Gradient descent and Bayesian optimization for lithium-ion battery aging diagnostics

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HIGHLIGHTS

- Machine learning enhances differential voltage analysis for battery diagnostics.
- Gradient descent offers fast but initially unstable curve fitting.
- Fitting quality and stability improves by selecting best fits from multiple runs.
- Bayesian optimization provides stable results but at much higher computational cost.

ABSTRACT

Accurate and efficient parameter estimation is essential for battery diagnostics and aging analysis. In this study, we compare two optimization-based approaches—gradient descent and Bayesian optimization—for extracting parameters from differential voltage analysis in lithium-ion batteries. While these techniques are widely used, their relative strengths and limitations for this application are not well understood. The study evaluates the trade-offs between these methods in terms of result quality, computational cost, and reliability within this specific application. The diagnostic results from our battery data suggest adopting gradient descent as an initial method for rapid and efficient analysis, while employing more stable optimization techniques, such as Bayesian optimization, as a verification step to mitigate potential instability. Comparing the two methods provides information on algorithmic choice, while inspiring further discussions on selecting appropriate techniques for specific research tasks.

1. Introduction

Data limitations, scalability, and accuracy are common challenges in materials research [1–3]. Gradient descent and Bayesian optimization are two promising approaches to overcome these common materials science challenges as both can automatically solve optimization problems effectively with small amounts of data [4–7]. However, there is a deficit in understanding how these two methods compare for specific tasks. Identifying the strengths and limitations of the two algorithms can provide insights for selecting the proper optimization method given different resource-limited scenarios. This paper benchmarks gradient descent and Bayesian optimization on a lithium-ion battery (LIB) aging diagnostic dataset and examines when each algorithm is the most effective and how to best utilize them based on specific research tasks. LIB aging diagnostics is an ideal benchmarking case to study because it is constrained by data and needs to be scaled without compromising accuracy.

As LIBs charge and discharge, their performance deteriorates over time due to various aging mechanisms [8,9]. The directly observable effects of LIB aging are categorized into two modes: capacity fade (i.e. inability to transport as many electrons during each charge/discharge cycle) and power fade (i.e. each cycle carries less energy per unit time) [10]. One direct and reliable way to analyze each aging mode is dismantling the cell and investigating the chemical changes (i.e., post-mortem analysis) [11,12]. However, due to the destructive nature of post-mortem analysis, many identical cells need to be prepared for different cycles. The consequences of having to prepare multiple identical cells are twofold: i) assembling many cells is time-consuming and costly, and ii) variances are introduced since no two cells are exactly the same.

Therefore, it is desirable to leave the cell intact after each cycle, and obtain performance information from each cell at different cycle numbers. One way to achieve non-invasive diagnostics is extracting LIB aging information through voltage (V) and capacity (Q) measurements

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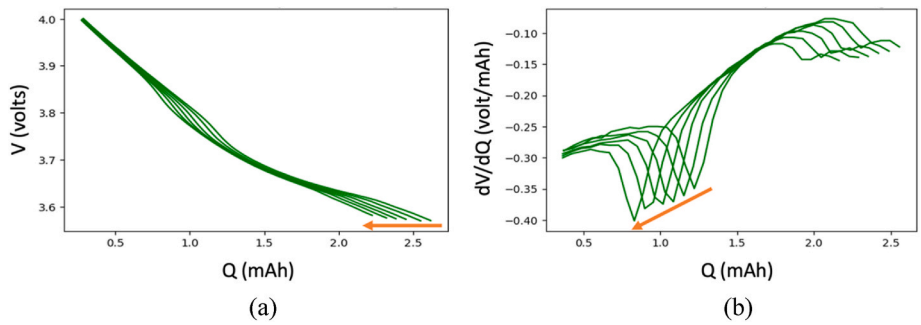

Fig. 1. Example of NMC532-Graphite cell discharge: (a) voltage curve and (b) differential voltage curve after different numbers of cycles. The arrows indicate direction of increasing cycle number.

[9,13–25], and one such example is differential voltage analysis (DVA) [14]. Fig. 1 illustrates DVA for a cell containing a $LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$ (NMC532) cathode and a graphite anode. Fig. 1 (a) shows deformation of the VQ curve as the cycle number increases. The deformation becomes more obvious in the differential voltage curves (Fig. 1 (b)) where the peaks in differential voltage curves move towards the left and intensify as cycle number increases. Although it is visually obvious that the differential voltage curves encode battery aging information, extracting quantitative values from the data is not a straightforward task and can be labor intensive if done manually. Therefore, there have been various attempts to apply machine learning methods to extract battery aging information [13,19–24].

The use of machine learning in battery research has been expanding and goes beyond analyzing battery aging. Since machine learning is a diverse family of algorithms, different methods can contribute to battery science across a large range of research areas. For example, Bayesian optimization and genetic algorithms have been used for inverse design of batteries [26]; random forest has been applied to analyzing coatings and optimizing manufacturing [27]; reinforcement learning has been used to develop fast charging strategies [28]; gradient descent combined with polynomial regression can monitor battery health [29]. Despite the variety of machine learning applications across battery research, a major challenge in applying machine learning to battery science is data scarcity [26]. In many cases we have limited experimental data despite needing many input features to build accurate predictive models, which can lead to overfitting and unreliable results. Additionally, interpretability can be another research challenge; while many models can accurately describe and predict battery behavior, they often are not easily interpreted by humans [27]. A final challenge in applying machine learning to battery research is the tendency to focus on a narrow range of algorithms, such as support vector machine and neural networks, instead of exploring a full range of machine learning algorithms [27]. This oversight can lead to improper application of machine learning methods and poor results.

Given the diversity of machine learning methods and the specific challenges faced in battery research, it becomes crucial to examine how different algorithms perform in particular applications. In this work, we focus on two algorithms—Bayesian optimization and gradient descent—applied to LIB aging diagnostics. A common issue in studies that apply machine learning to LIB aging and other materials research is

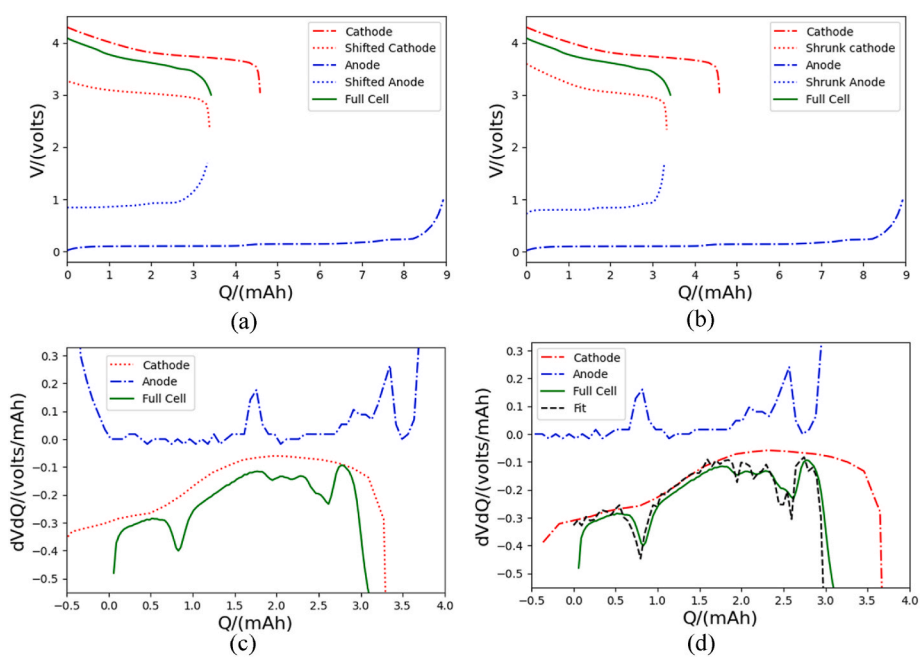

Fig. 2. Example of deforming DVQ and VQ curves. (a) shifted cathode and anode half cells (dash dotted curve) compared to the original half (dotted curves) and full cell voltage curve (solid green curve). (b) shrunk cathode and anode half cells (dash dotted curve) compared to the original half (dotted curves) and full cell voltage curve (solid green curve). (c) Example of original full cell (solid green curve) and half-cell differential voltage curves (dashed blue and red curves). (d) Example of deforming half-cell differential voltage curves to fit (black dashed) full cell experimental (solid green curve) differential voltage curves. The dashed curves in (c) and (d) are vertically offset from dot dashed curves for clearer visualization. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

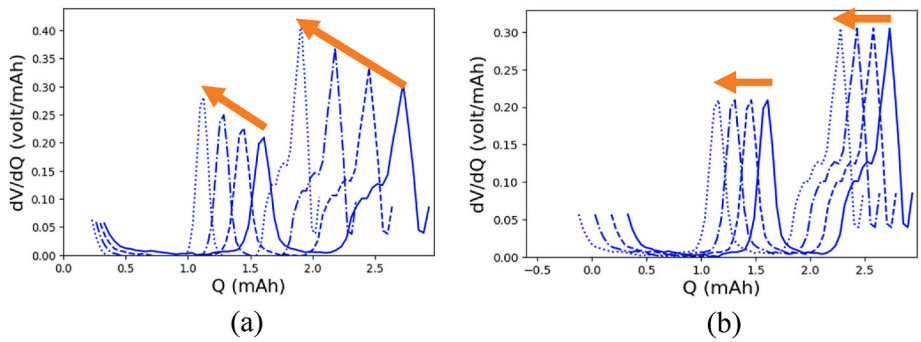

Fig. 3. Examples of graphite half-cell differential voltage curves that (a) shrink and (b) shift after cycling. The arrows indicate direction of increasing cycle number.

the lack of explanation for why a particular algorithm is chosen. Machine learning refers to a large and diverse family of algorithms that have unique advantages and disadvantages [30]. Machine learning algorithms and methods are not universally applicable to all types of problems; different problems typically require different algorithms for optimal performance [31]. The selection of appropriate algorithms for different research tasks in materials research is an important topic, which is often overlooked in the literature. There are two approaches to address the gap. One approach surveys a broad range of machine learning algorithms [31–33]. The other approach uses a concrete example to benchmark performance of different algorithms. We believe the benchmarking approach is under-represented in literature, and deserves more attention because machine learning performance is highly task-specific and can hardly be thoroughly explained without concrete examples or benchmarking cases [31]. While some existing papers directly compare different machine learning algorithms on benchmarking tasks [34–36], few of them compare the performance of gradient descent and Bayesian optimization [37,38], which are two important algorithms in the machine learning family for optimization problems in materials research [7,39–42]. However, important gaps remain. For example [37] only compare final results, leaving out important details on other important factors. Algorithm selection should not only be based on final performance. Computational time, stability, etc. should also be considered [38]. provides a more detailed comparison of Bayesian optimization and gradient-based methods (e.g., L-BFGS-B) in the context of nano-optics, reporting that Bayesian optimization outperforms gradient-based approaches in both objective performance and runtime. While our study follows a similar comparative approach, it focuses on a different application domain (LIB aging diagnostics) and arrives at notably different conclusions.

Our findings reveal that gradient descent can exhibit instability and high sensitivity to initialization, which was not emphasized in Ref. [38]. Furthermore, we highlight that both Bayesian optimization and gradient descent have complementary strengths, depending on problem characteristics such as data noise, initialization sensitivity, and interpretability. This suggests that the relative performance of these methods is highly application-dependent, and that more nuanced, domain-specific benchmarks are essential when selecting an optimization strategy.

The following sections briefly introduce the DVA method, our experimental setup, gradient descent, and Bayesian optimization methods. The methods are then used to study LIB aging diagnostics, and their results are shown and compared, highlighting the differences of these two algorithms in performance, computational costs, and potential usage scenarios.

2. Differential voltage analysis

DVA is a common approach for non-destructive analysis of cell degradation. Both electrodes contribute to the shape and peaks of a full cell differential voltage curve, and it can be difficult to identify which electrode contributes to the aging for different mechanisms [43]. To address this challenge, methods using reference cathode and anode half-cell differential voltage curves have been developed. Reference half-cell differential voltage curves can be deformed through shifting and shrinking the reference data to fit cycled full cell differential voltage curves to parse out contributions from each electrode. Spline fitting is used to smooth the data and reduce noise during differentiation.

An example of shifting and shrinking pristine half-cell differential voltage curves to fit a full cell curve is shown in Fig. 2. Fig. 2 (a) and (b) show experimental pristine half-cell voltage curves for the anode and cathode, and cycled full cell voltage curves. In addition, the figures also show shifted and shrunk pristine half-cell voltage curves. As the effects of shifting and shrinking are difficult to visualize, Fig. 2 (c) shows the original differential voltage curves of pristine half-cell and full cell, while Fig. 2 (d) shows the combined effect of shifting and shrinking of pristine half-cell differential voltage curves to reconstruct the full cell differential voltage curve. Based on the shifting and shrinking of the curve, cell aging information can be quantitatively inferred [44]. The mathematical details of the reconstruction (or fitting) process are described in the Methods section of this paper.

For each half-cell, deformation of the differential voltage curves is algebraically caused by two factors, namely shrinkage and shift. Fig. 3 shows the geometric effects of the two factors. The first factor is the shrinkage, denoted by k, indicating that the differential voltage curve shrinks by a fraction k. Geometrically, shrinkage causes both peak positions and width of the differential voltage curve to shrink towards the left, intensifying the peaks. The second factor is the shift, denoted by b, indicating that the differential voltage curve moves to the left by b mAh. Geometrically, the shift causes differential voltage curves to shift uniformly towards the left, leaving the peak intensities unchanged. Physically, in each half-cell, k is associated with a loss of active material (LAM), indicating that a fraction of the active material remains in the half-cell; b is associated with a loss of lithium inventory (LLI), indicating that b mAh of cyclable lithium ion is lost [9].

3. Experimental materials and methods

The positive and negative electrodes used in this study were obtained from the Cell Analysis, Modeling and Prototyping (CAMP) facility at Argonne National Laboratory. The positive electrode comprised a thick coating with 90 wt% $LiNi_{0.5}Mn_{0.3}Co_{0.2}O_2$ (NMC532), 5 wt% C45 carbon conductive additive and 5 wt% polyvinylidene fluoride (PVdF) on a 20 μm thick Al current collector. The negative electrode comprised a coating with ~92 wt% graphite, 2 wt% C45 carbon and 6 wt% PVDF binder, on a 10 μm thick Cu current collector. The electrolyte was a solution of 1.2 M lithium hexafluorophosphate ($LiPF_6$) salt dissolved in a 3:7 w/w solvent mixture of ethylene carbonate (EC) and ethyl methyl carbonate (EMC) and the separator was Celgard 2325. Before cell assembly, the electrodes and separator were dried in vacuum ovens set at 120 °C and 70 °C, respectively. All electrochemical experiments were conducted in 2032-type coin cells (1.54 cm^2 area electrodes) assembled in an Ar-atmosphere glove box and galvanostatically cycled within a

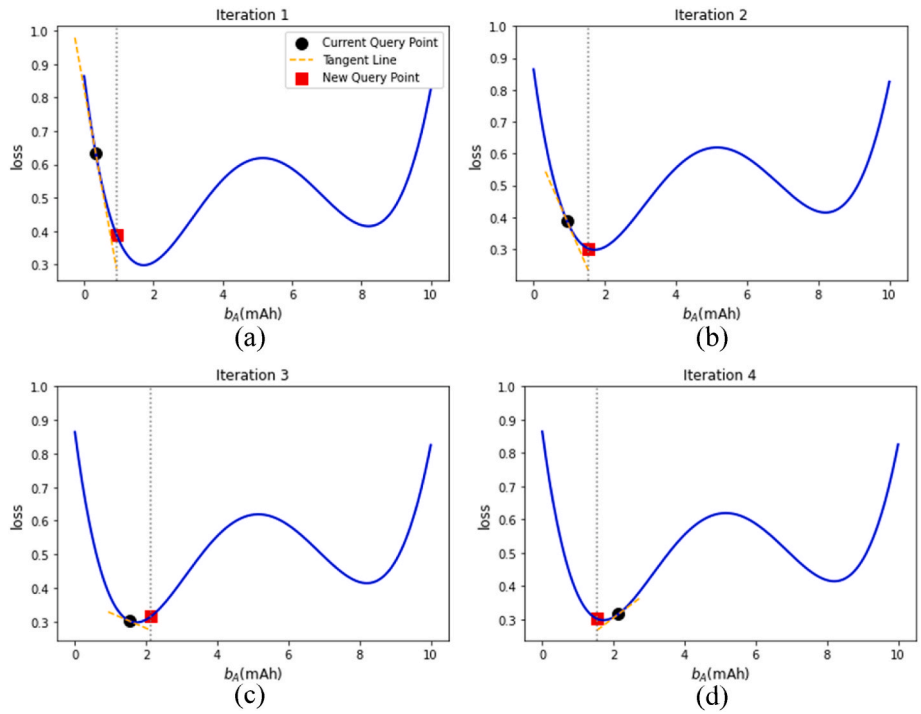

Fig. 4. Visualization of gradient descent finding the best b_A value that minimizes the loss after the (a) first, (b) second, (c) third, and (d) fourth iterations. The blue curve is the loss function. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

constant temperature oven held at 30 °C. The positive electrode half-cell was cycled in the 2.5–4.3 V range vs. Li/Li^+, whereas the negative electrode half-cell was cycled in the 1.5–0.01 V range vs. Li/Li^+. The full cell, after initial formation cycles, was cycled between 3 and 4.1V at C/3 rate, with periodic reference performance measurements at C/25 rate. All electrochemical tests were conducted using a Maccor 2400 cycler.

4. Gradient descent and Bayesian optimization

Gradient descent and Bayesian optimization are two fundamentally different approaches to optimization: the former takes a local approach while the latter takes a global approach. Due to the fundamental differences, they offer complementary strengths. In this section, a brief explanation of their differences and strengths are given, using LIB diagnostics as an example.

The benchmarking case in this research utilizes voltage (V) and capacity (Q) information taken at different cycle numbers throughout the battery lifetime to diagnose LIB aging by quantifying the amount of shift (LLI) and shrinkage (LAM) of cycled cell differential voltage curves compared to pristine half-cell differential voltage curves. To quantify the shift and shrinkage in half-cells, four parameters k_A, k_C, b_A, and b_C are defined. Parameter k quantifies shrinkage of the differential voltage curve as a fraction k; parameter b quantifies shift of the differential voltage curve by b mAh to the left. Subscript A stands for anode half-cell while C stands for cathode half-cell. Figs. 2 and 3 visualize the effects of k and b on both differential voltage curves and voltage curves.

The voltage curve of a cycled half-cell ($\widetilde{V}_i$ (q)) is deformed by k and b from the curve of a pristine cell (V_i (q)), where the subscript i can take values A for anode and C for cathode. When the deformed voltage curve has shifted from the original, it can be defined as,

$$\widetilde{V}_i(\mathrm{q}) = V_i(\mathrm{q} + b_i) \tag{1}$$

where q is the capacity, or when the deformed voltage curve has shrunk as, $\widetilde{V}_i(q) = V_i\left(\frac{q}{k_i}\right)$. (2)

Taking derivatives on both sides of Eqs. (1) and (2

$$\frac{d\widetilde{V}_i}{\mathrm{d}Q}\Big|_q = \frac{dV_i}{dQ}\Big|_{\mathrm{q}+b_i} \tag{3}$$

$$\frac{d\widetilde{V}_i}{\mathrm{d}Q}\Big|_q = \frac{1}{k_i}\frac{dV_i}{dQ}\Big|_{q/k_i} \tag{4}$$

Combining shift and shrinkage in the anode, we have

$$\frac{d\widetilde{V}_A}{\mathrm{d}Q}\Big|_q = \frac{1}{k_A}\frac{dV_A}{dQ}\Big|_{(\mathrm{q}+\mathrm{b_A})/k_A} \tag{5}$$

where b_A and k_A is the shift and shrinkage of the anode, respectively. Similarly, let b_C and k_C be the shift and the shrinkage of cathode, respectively,

$$\frac{d\widetilde{V}_C}{\mathrm{d}Q}\Big|_q = \frac{1}{k_C}\frac{dV_C}{dQ}\Big|_{(\mathrm{q}+\mathrm{b_C})/k_C} \tag{6}$$

Equations (1)–(6) cannot directly diagnose LIB full cells, since voltages are not measured separately for the anode and cathode in a full cell. Instead, the full cell voltage V_F is measured as a whole, and can be written as

$$V_F = V_C - V_A \tag{7}$$

Taking the derivative on both sides of Equation (7) we have

$$\frac{dV_F}{\mathrm{d}Q} = \frac{dV_C}{\mathrm{d}Q} - \frac{dV_A}{\mathrm{d}Q} \tag{8}$$

To diagnose a cycled full cell, we can define an artificial full cell differential voltage curve ($\frac{d\widetilde{V}_F}{\mathrm{d}Q}$) as

$$\frac{d\widetilde{V}_F}{\mathrm{d}Q} = \frac{d\widetilde{V}_C}{\mathrm{d}Q} - \frac{d\widetilde{V}_A}{\mathrm{d}Q} \tag{9}$$

From Eqs. (5) and (6), by varying parameters b_A, b_C, k_A, and k_C, the artificial cycled full cell differential voltage curve $\frac{d\widetilde{V}_F}{\mathrm{d}Q}$ can be reshaped. When the artificial full cell $\frac{d\widetilde{V}_F}{\mathrm{d}Q}$ agrees well with an experimental $\frac{dV_F}{\mathrm{d}Q}$, the

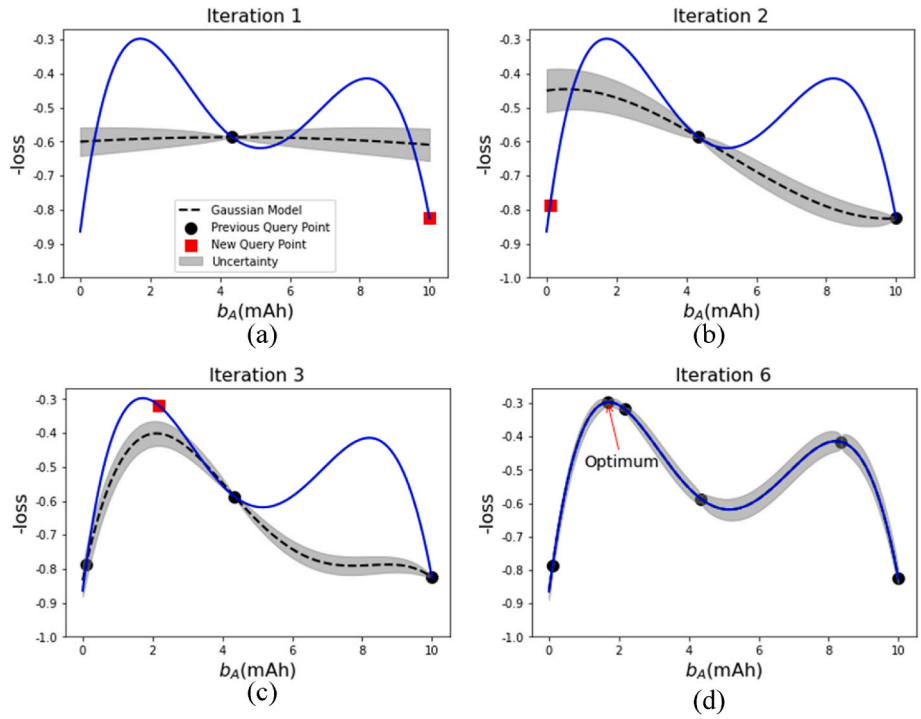

Fig. 5. Visualization of Bayesian optimization finding the best b_A value that maximizes the *-loss* after the (a) first, (b) second, (c) third, and (d) sixth iterations. The blue curve is the -loss function. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

corresponding b_i and k_i can be used to compute the LLI and LAM of the anode and cathode respectively.

Mathematically, the goal is to determine a set of b_A, b_C, k_A, and k_C values such that it minimizes

$$loss = 1\Big/n\left\|\frac{dV_F}{dQ} - \frac{d\widetilde{V}_F}{dQ}\right\|_{L_2} \tag{10}$$

where n is the length of the $\frac{dV_F}{dQ} - \frac{d\widetilde{V}_F}{dQ}$ vector, and $\|\cdot\|_{L_2}$ indicates the L_2 norm. Or equivalently, the goal is maximizing *-loss* with a set of b_A, b_C, k_A, and k_C values. Once the *loss* function is minimized (or the *–loss* function is maximized), the set of b_A, b_C, k_A, and k_C values can be used to compute LLI and LAM for each electrode in the full cell. In a simplified explanation, the quantity of LAM for each electrode is $1 - k_i$, and the maximum value of b_i represents an estimate of the total LLI. For a more comprehensive analysis, refer to Birkl et al. [9].

In order to use k_i and b_i to minimize the *loss* function automatically, gradient descent and Bayesian optimization were used. Gradient descent minimizes the *loss* function and Bayesian optimization maximizes the *–loss* function.

4.1. Gradient descent

For visualization, Fig. 4 shows an example of how gradient descent finds the best anode shift b_A to minimize the *loss* function. The actual value of *loss* at each b_A is depicted by the solid blue curve and is unknown. Gradient descent starts by selecting a random b_A value (the current query point, represented by the black dot in Fig. 4 (a)) and evaluates the *loss*. At the current query point, the numerical gradient is computed (represented by the orange dashed line tangent to the curve at the current query point). Based on the gradient, a new query point (represented by the red square in Fig. 4 (a)) is picked to evaluate by moving a small amount towards the opposite direction of the gradient. Mathematically,

$$b_A^{new} = b_A^{current} - \alpha\nabla(loss) \tag{11}$$

where b_A^{new} is the new query point where the *loss* function is evaluated, $b_A^{current}$ is the current query point, α is a small numerical number called the learning rate that can either be set as a constant or adaptive, and $\nabla(loss)$ is the numerical gradient at $b_A^{current}$ [45,46]. Equation (11) is computed iteratively until the b_A^{new} value converges. Note that gradient descent only uses local gradient information for optimization in each iteration. As a result, it is typically computationally inexpensive, but at the cost of possibly getting trapped at a local minimum [37].

Additionally, gradient descent relies on gradients of the objective function, and when working with limited data, estimating these gradients accurately can be challenging—particularly if the function is noisy or learned from data (e.g., via a surrogate model) [4,47]. In such cases, the optimization may become sensitive to noise or prone to instability, which can resemble overfitting in practice.

However, gradient descent itself does not inherently cause overfitting; rather, overfitting arises from imperfections in the model or limitations in the data to which gradient descent is applied. To mitigate these issues, regularization techniques, noise smoothing, and uncertainty-aware modeling approaches are commonly used. In our work, we employed spline interpolation to smooth the function, which helped reduce the influence of noise and prevent unphysical behavior in the resulting model.

4.2. Bayesian optimization

Fig. 5 shows an example of how Bayesian optimization finds the best anode shift b_A to maximize the *–loss* function. The actual value of *–loss* at each b_A is depicted by the solid blue curve in Fig. 5 and is unknown. Bayesian optimization starts by selecting a random b_A value and evaluates the *–loss* (represented by the black dot in Fig. 5 (a)). A Gaussian model can be built based on the explored b_A value and evaluated *–loss*. The black dashed curve indicates the Gaussian mean and the grey area indicates the uncertainty of our current model. Uncertainty of the Gaussian model increases as the b_A value moves further away from explored points. Based on the Gaussian model, Bayesian optimization picks another b_A value that maximizes the *-loss* expectation (represented

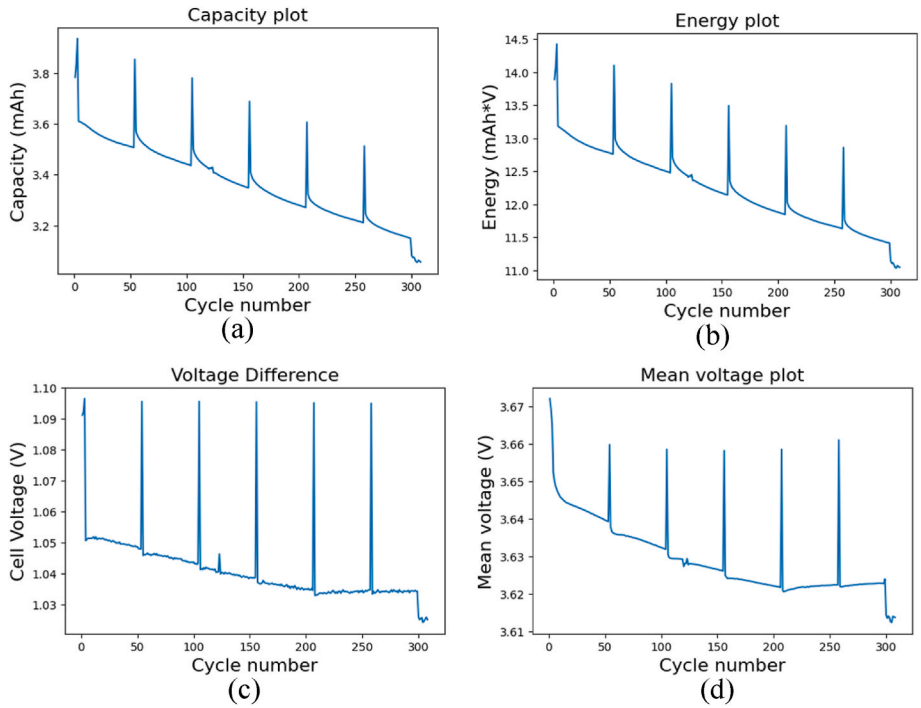

Fig. 6. NMC532-Graphite cell plots as a function of cycle number. (a) capacity, (b) total energy, (c) voltage differences between beginning and end of a cycle, and (d) mean voltage after different numbers of cycles.

by the red square at the end of the curve in Fig. 5 (a)) and evaluates the *–loss* at the selected b_A value (represented by the second black dot at the end of the curve in Fig. 5 (b)).

Based on the two points explored, Bayesian optimization updates the Gaussian model to improve predictions of the *–loss* (represented by the updated black curve and grey area in Fig. 5 (c)) based on a preset policy called the acquisition function [4]. After an adequate number of iterations of the exploration, evaluation, and updating process, Bayesian optimization picks the best b_A value (pointed out by the arrow in Fig. 5 (d)) which maximizes the *–loss*, representing the best anode shift to fit the experimental data. In contrast to gradient descent, Bayesian optimization updates a global model instead of using only local information in each iteration. As a result, Bayesian optimization is typically more computationally expensive, but less prone to being trapped at a local minimum compared to gradient descent.

5. Implementation

To compare the Bayesian optimization and gradient descent approaches to battery diagnostics, data from a NMC532-Graphite cell was studied. The cell was cycled for 309 cycles with a C/25 rate for cycles 54, 105, 156, 207, 258, and 309, and a C/3 rate for the rest of cycles. The voltage and capacity information in cycles 54, 105, 156, 207, 258, and 309 at the discharging step was used for analysis. For each of the six cycles, the data were used to determine the derivative of the voltage with respect to capacity, i.e. the $\frac{dV_F}{dQ}$ was computed for each data point.

Two half-cells, NMC532-Li and Graphite-Li, were used as the reference cells. The differential voltage plots from the NMC532-Li cell was used as the cathode $\frac{dV_C}{dQ}$; information from the Graphite-Li cell was used as the anode $\frac{dV_A}{dQ}$. Using Equations (5) and (6), Bayesian optimization and gradient descent varied b_A, b_C, k_A, and k_C to find the DVQ curve of an artificial full cell (Equation (9)) to maximize the *–loss* (Equation (10)).

Since the reference half-cells and full cell did not measure voltage information at exactly the same instantaneous capacity, to compute the *–loss* it was necessary to mesh the x-axis (i.e. the capacity axis) using the x-axes of deformed half-cell differential voltage curves ($\frac{d\tilde{V}_C}{dQ}$ and $\frac{d\tilde{V}_A}{dQ}$) and full-cell differential voltage curve ($\frac{dV_F}{dQ}$). Namely, the maximum of the left end point (x_1) and minimum of the right end point (x_2) of $\frac{d\tilde{V}_C}{dQ}$, $\frac{d\tilde{V}_A}{dQ}$, and $\frac{dV_F}{dQ}$ x-axes were selected, and all x values of deformed half-cells and full cells between x_1 and x_2 were selected to create a new sample collection of x values. The mesh allows interpolation of all differential voltage curves. In addition to making it possible to compute the *loss*, the interpolation also smoothed the differential voltage curves to mitigate the noise in the measurement and differentiation.

The Bayesian optimization started with 40 random initial exploration points and ran 700 iterations. Each C/25 cycle was fitted with 3 different acquisition functions (expected improvement, possibility of improvement, and upper confidence bound) [4], and each acquisition function was run 3 times with different random initial 40 exploration points. The gradient descent was set to run 9 times for each cycle to match the Bayesian optimization. Each time the gradient descent started with a different random starting point, used the limited-memory BFGS optimization method, and was allowed a maximum of 5000 iterations.

6. Results and discussions

As shown in Fig. 6 (a), our NMC532-Graphite cell lost 0.43 mAh capacity as it cycled from cycle 3 to cycle 309. The total energy (Fig. 6 (b)) showed a similar trend as the capacity. The voltage differences (Fig. 6 (c)) between the beginning of discharge and end of discharge had negligible changes. Additionally, the average voltage did not change more than 1 %. As the voltage stayed stable, it can be concluded that there was minimal power fade. As the total amount of energy storage decreased and more importantly the capacity decreased, it can be concluded that the major mode of aging is capacity fade. However, based on only the information in Fig. 6, it is difficult to quantify how much LLI and LAM contributed to the total capacity loss.

To diagnose the LLI and LAM contributions, DVA with gradient descent and Bayesian optimization were used. First the experimental differential voltage curves were fit to the pristine half-cell differential

Table 1
Comparison of run time and performance of Gradient Descent and Bayesian Optimization.

	Gradient Descent	Bayesian Optimization (Run 1)	Bayesian Optimization (Run 2)
Run time	128 s	4442 s	16148 s
Mean MSE	0.024	0.00087	0.00088
Standard deviation of mean MSE	0.17	0.00045	0.00045
Average min MSE	0.00049	0.00055	0.00055
Mean MAE	0.0478	0.0215	0.0215
Mean percentage absolute error	9 %	8.9 %	8.9 %

voltage curves. The fitting relied on the fact that the full cell voltage is the difference between the cathode voltage and anode voltage, hence the full cell differential voltage can be expressed as the difference between the cathode differential voltage and anode differential voltage [25]. Therefore, NMC532-Li and Li-Graphite half-cells discharging differential voltage curves were shifted and shrunk to fit the cycled NMC532-Graphite full cell differential voltage curves after cycles 54, 105, 156, 207, 258, and 309 using gradient descent and Bayesian optimization.

As a comparison of the performance of gradient descent and Bayesian optimization, the average run time for each fitting was 128 s for gradient descent and 4442 s for Bayesian optimization as shown in Table 1. In other words, Bayesian optimization ran 40 times slower than gradient descent. The higher computational cost has the tradeoff of higher stability. In around 9 % (14 out of 162) of the runs, gradient descent could not converge due to poor selection of a starting point. Bayesian optimization, on the other hand, reliably converged in each case. Even in the converged cases, gradient descent on average performed worse than Bayesian optimization. We calculated the mean square error (MSE), mean absolute error (MAE), and mean percentage absolute error over 6 cycles and 9 iterations for each cycle. The MSE of gradient descent in the converging cases was 0.024 with a standard deviation 0.17. In comparison, the MSE of Bayesian optimization was only 0.00087 with a standard deviation 0.00045, as shown in Table 1. Example cases for both are shown in Fig. 7.

However, if only the best fit (fit with the least *loss*) of each cycle was chosen, the MSE of all cycles for gradient descent was 0.00049, better than Bayesian optimization which was 0.00055. Fig. 8 shows the quantitative diagnostic results of best-fitting cases of each cycle for b_A, b_C, k_A, and k_C. Both share the same trend that LAM and LLM increase in both the anode and cathode, which is physically and chemically plausible. However, when run the same number of iterations as gradient descent (blue disks in Fig. 8), Bayesian optimization shows a random fluctuation in the best fitting cases (orange squares in Fig. 8). After running 4 times as many iterations (36 iterations for each cycle) as gradient descent, Bayesian optimization best fittings (red triangle in Fig. 8) converged to gradient descent results but were still not as physically plausible as gradient descent, with a higher minimum MSE of 0.00053. The better fitting of gradient descent demonstrates that it has lower computational cost and can potentially pick up physical information more effectively in the case of diagnosing the aging of our sample cell, although it is not as stable as Bayesian optimization.

Fig. 8 (b) and (d) indicate that the anode and cathode are gradually losing capacity based on the almost linear increase of b_A and b_C. From cycle 54 to cycle 309, the changes in b_A and b_C are both approximately 0.6 mAh in the anode and cathode, which validates the LLI result. Fig. 8 (a) and (c) indicate that the anode and cathode both lost active materials. The cathode lost approximately 6 % of active material relative to the pristine cathode, corresponding to 0.252 mAh of capacity. The anode lost approximately 2.5 % of active material relative to the pristine anode, corresponding to 0.135 mAh of capacity. Since the cathode is the capacity-limiting electrode in this system, we assume that most of the observed LLI originates from processes occurring at the cathode. To better understand the other mechanisms contributing to LLI, such as parasitic reactions or solid electrolyte interphase (SEI) growth, we quantified the fraction of LLI attributable to LAM at the cathode.

Fig. 8(e) shows the analysis. We observe a sharp increase in LAM cathode contribution to LLI immediately after cycle 105, followed by a

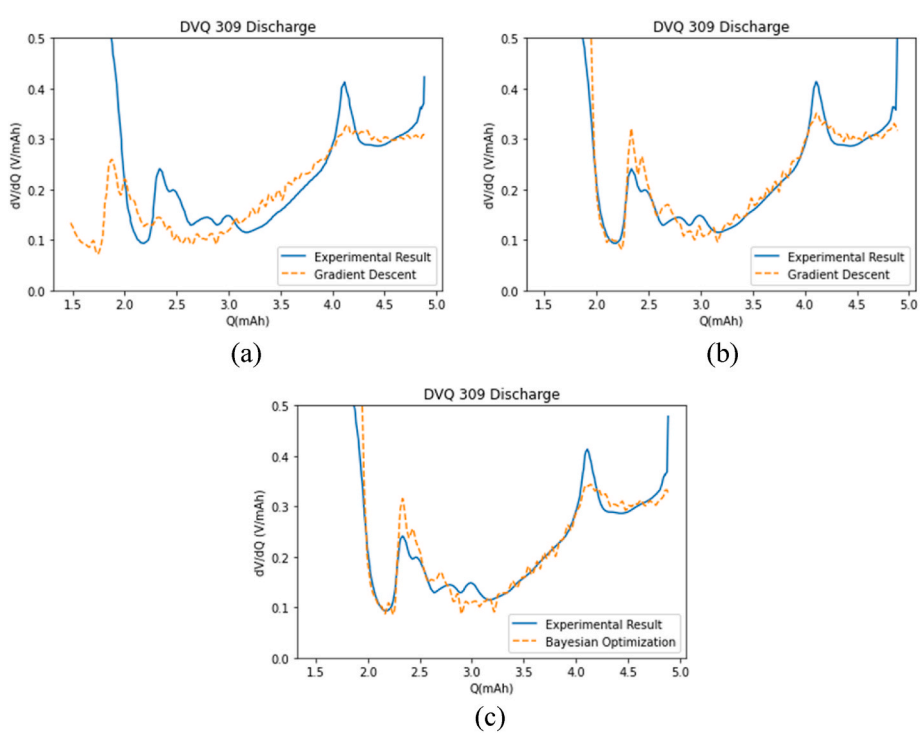

Fig. 7. Examples of gradient descent and Bayesian optimization fittings. Solid curves are ground truths (experimental results) and dashed curves are fitting results. (a) Failed (divergent) case of gradient descent fitting. (b) Successful (convergent) case of gradient descent fitting. (c) Bayesian optimization fitting.

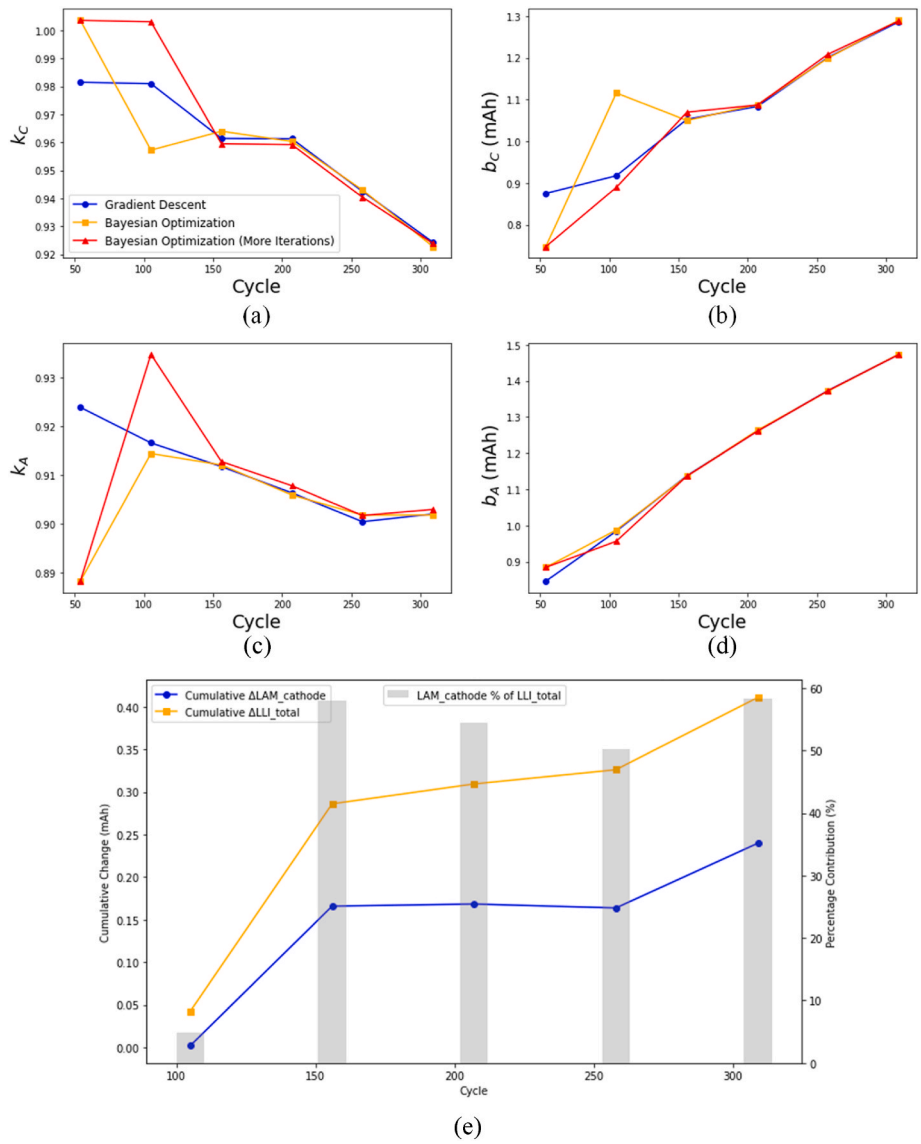

Fig. 8. Quantitative diagnostic results of (a) k_C, (b) b_C, (c) k_A, and (d) b_A. The gradient descent results are in blue dots. The Bayesian optimization results with the same number of iterations are in orange squares, and Bayesian optimization results with 4 times as many runs as gradient descent are in red triangles. (e) shows the cumulative LLI and LAM cathode, along with their percentage breakdown. Blue dots represent the cumulative LAM cathode, yellow squares represent the cumulative LLI, and the grey histogram indicates the percentage of LAM contributing to LLI. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

plateau. This increase of percentage contribution of LAM cathode in LLI suggests that critical degradation events in cathode may have occurred between cycle 105 and 156, and warrant further investigation.

By the end of the test, approximately 60 % of the total capacity loss was attributed to LAM at the cathode, making it the dominant degradation mechanism. The remaining 40 % was due to other sources of LLI. The result underscores the critical need to enhance the structural stability of the NMC532 cathode to mitigate LAM-driven capacity loss. At the same time, understanding and addressing additional contributors to LLI—such as SEI growth, lithium trapping, and parasitic side reactions—remains important for improving overall cell longevity.

7. Conclusions

In this research we demonstrated how machine learning methods can be applied to examine performance loss mechanisms during the aging of an NMC532-Graphite cell. By applying optimization techniques, commonly used in machine learning, to differential voltage analysis, we were able to track how key parameters evolved with cycling. Notably, the analysis revealed a rapid increase in LAM at the cathode after cycle 105. These findings point to specific time windows that warrant further physico-chemical investigation and highlight the significant role of cathode degradation in overall capacity loss for this system.

To carry out the fitting of experimental differential voltage curves, we compared two optimization strategies: gradient descent and Bayesian optimization. These methods were evaluated in terms of fitting accuracy, computational cost, and robustness. The comparisons revealed notable differences between the two approaches. While gradient descent is much less stable than Bayesian optimization, it requires less than 2.5 % of the computational cost of Bayesian optimization. Meanwhile, the best fitting cases in gradient descent had lower losses and fittings with less fluctuations. In other words, the instability of gradient descent can be mitigated by running multiple fittings with different starting points and picking the best-performing fitting. The results suggest that as a practical approach in future materials studies, gradient descent should be implemented first, followed by Bayesian

optimization or other computationally expensive yet stable machine learning algorithms to verify its stability and reliability. If the results of both gradient descent and Bayesian optimization align closely, we can rely on the gradient descent output. If not, it indicates a need for more computationally intensive methods. In the case presented in this paper, the results were consistent (Fig. 8), so no further computation was necessary.

This work focused on comparing gradient descent and Bayesian optimization on LIB diagnostics. Overall, more machine learning and optimization methods need to be studied by the materials science community and benchmarked; these include methods such as genetic algorithms and simulated annealing. This paper only considers optimization problems. Regression is another important branch of tasks in materials research that can benefit from machine learning algorithms. It is critical to both academia and industry to understand when and how different methods can be applied to research tasks, and to benchmark the methods such as kernel ridge regression, random forest, and different architectures of neural networks in materials research.

CRediT authorship contribution statement

Ziqing Zhao: Writing – review & editing, Writing – original draft, Visualization, Software, Methodology, Formal analysis, Conceptualization. **Joseph Kubal:** Writing – review & editing, Validation, Supervision, Software, Methodology, Formal analysis, Conceptualization. **Daniel Abraham:** Writing – review & editing, Supervision, Resources, Formal analysis, Data curation. **Emily Ryan:** Writing – review & editing, Supervision, Funding acquisition.

Software Availability

Experimental data for training and validating the DVA + Bayesian optimization was obtained from CAMP at Argonne National Laboratory. The data presented in this work was obtained through Python scripts implementing the DVA + Bayesian optimization model. All codes are available on GitHub, https://github.com/emilymryan.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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